



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 -Electricity Markets and Energy System Planning, Specialisation Course

An Optimisation Model for the Electricity Market Clearing

Hossein Farahmand

Professor

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Agenda

- Course purpose and learning journey
- Teaching team and practical information
- Project workflow, milestones and supervision
- Today's recap: social welfare, prices and duality
- Active checkpoints and final learning outcomes

Today's learning outcomes

By the end of this lecture, you should be able to:

Explain

centralized market clearing and maximises social welfare

Formulate

a simple pool-trading market-clearing LP

Interpret

the power-balance dual variable as the clearing price

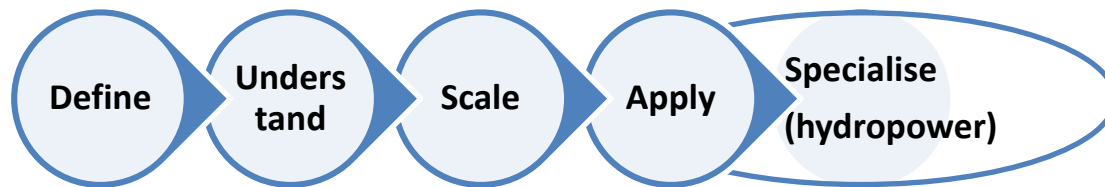
Connect

primal, KKT and dual views before moving to uncertainty

Learning question for today: How can one optimisation problem produce both accepted quantities and the market price?

Goals

- Formulate energy market and production planning optimisation problems under uncertainty.
- Develop an intuition for how and why optimal decisions under uncertainty differ from decisions made using deterministic scenarios.
- Know how large optimisation problems under uncertainty can be solved using decomposition methods.
- Give students practical experience with writing and structuring optimisation problems, as well as experience with data handling.
- Provide knowledge about optimisation techniques and models successfully applied to hydropower scheduling, from global to local models, from long- to short-term.



The Teaching Team



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Gro Klæboe

- Lecturer
- Adjunct Associate Professor, Dept. of Electric Energy, NTNU & Head of Nordic Analysis at Volve Insight
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- Scientific Assistant
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This course has two modules

	Flexibility in power grid operation and planning	Power market optimisation under uncertainty, decomposition and hydro power planning
Lectures	Mondays 12:15-14:00, Room: GL-SB1 S1	Fridays 10:15-12:00, Room: Gamle elektro EL2
Assessment	Project	Project
Main lecturer(s)	Iver Bakken Sperstad	Hossein Farahmand
Other	Common lectures with TET4575	

Lecture schedule

Time: Fridays 10:15–12:00

Venue: GL-GEL EL2

Week	Date	Session title	Lecturer
37	11.09.2026	Recap: An optimisation model for electricity market clearing	Farahmand
38	18.09.2026	Stochastic market clearing (two-stage stochastic programming)	Farahmand
39	25.09.2026	Tutorial 1: Stochastic optimisation formulation	Thorvaldsen
40	02.10.2026	Risk management for multi-market bidding	Farahmand
41	09.10.2026	Decomposition techniques	Farahmand
42	16.10.2026	Tutorial 2: Benders decomposition	Thorvaldsen
43	23.10.2026	Multi-stage stochastic programming	Farahmand
44	28.10.2026	Tutorial 3: Stochastic Dynamic Programming*	Thorvaldsen
44	30.10.2026	Scheduling hierarchy (long-term/seasonal/short-term) & water values	Farahmand
46	13.11.2026	Long-term scheduling - the EMPS model & area optimisation	Helseth
47	20.11.2026	Seasonal scheduling and coupling & short-term scheduling	Klæboe

* This tutorial will be held during the assignment supervision time on Wednesday 14:15–16:00 in S4 Sentralbygg 1.

Module assessment: Project with Oral presentation



- **In the project you will:**
 1. Present and motivate an optimisation problem
 2. Formulate the mathematical model
 3. Implement and solve it in Python/Pyomo
 4. Decompose the problem
 5. Analyse results and discuss assumptions
- Hydro power case is given, but we prefer that you formulate your own problem
- Synergies with project thesis is encouraged!
- **Problem requirements:**
 - includes uncertainty and recourse
 - can be written as a two-stage linear problem
 - is scoped small enough to finish well

Project: Hand in deadlines

Assignment supervision:

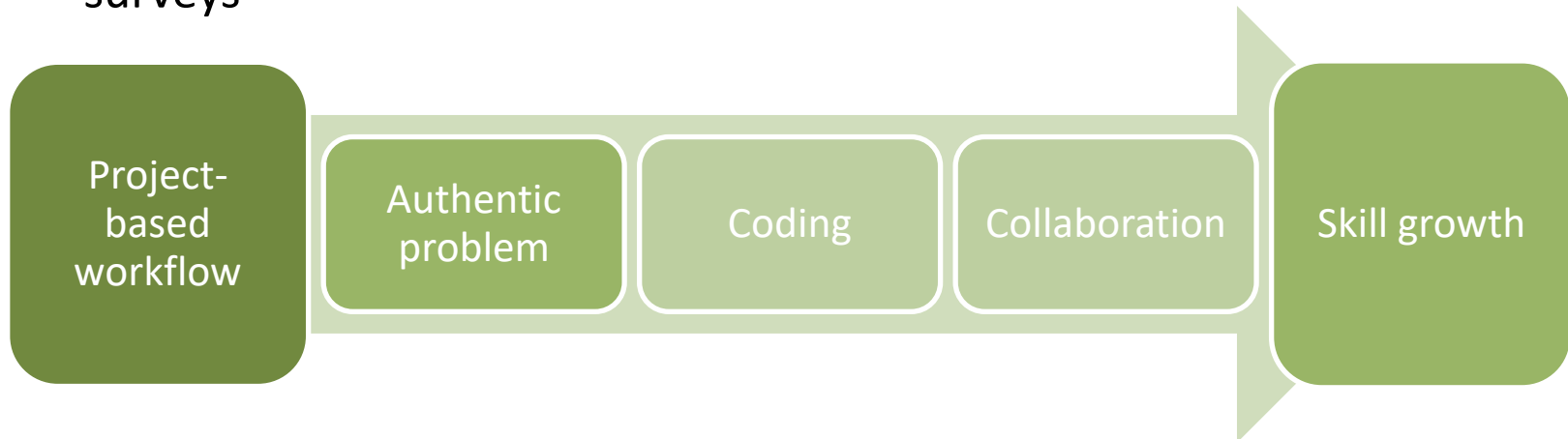
- Time: Wednesdays 14:15-16:00
- First supervision session: 16 September 2026 (week 38)
- Virtual supervision sessions in weeks 43 and 44
- Venue: S4 Sentralbygg 1
 - Map: <https://link.mazemap.com/L8nPzfl2>

Deadline	Week	Project part due
2026-09-25	39	Interim submission: Mathematical & textual problem formulation & data cleaning
2026-11-20	48	Full project Including part 4: Solving a two-stage problem using stochastic dynamic programming

- Use the milestones to avoid a large end-of-semester bottleneck.
- Interim submission is not formally graded, but they guide supervision.
- Bring concrete questions to tutorials: formulation, data, implementation, interpretation.
- Final submission: 20 November + oral project presentation (November 25 & 26).

Active Learning in TET4565 – Your Learning Journey

- Project-based workflow: Go from problem formulation to solving with advanced methods.
- Tackle authentic challenges: Focus on practical, real-world problems (Select the problem carefully and avoid overcomplicating the problem)
- Build coding skills: Gain hands-on experience in Python/Pyomo with continuous feedback.
- Learn together: Join interactive tutorials and collaborate with peers.
- Grow your expertise: Skill growth evidenced by student feedback and surveys



Overview (Recap)

- Centralized social welfare optimization
- How to derive Lagrangian function
- How to derive optimality conditions
- How to derive dual problem

Market clearing (Pool Trading): a simple example

Let us get started with a question. Assume an electricity market with two generators (G1 and G2) and two elastic demands (D1 and D2). What are the market-clearing outcomes (production, consumption and market-clearing price)?



Capacity: 50 MW
Offer price: €40/MWh



Capacity: 100 MW
Offer price: €100/MWh



Maximum load: 100 MW
Bid price: €150/MWh



Maximum load: 100 MW
Bid price: €50/MWh

Market outcomes:

- Productions of G1 and G2: ?
- Consumptions of D1 and D2: ?
- Market-clearing price: ?

Market clearing: a simple example

Let us get started with a question. Assume an electricity market with two generators (G1 and G2) and two elastic demands (D1 and D2). What are the market-clearing outcomes (production, consumption and market-clearing price)?



Capacity: 50 MW
Offer price: €40/MWh



Capacity: 100 MW
Offer price: €100/MWh



Maximum load: 100 MW
Bid price: €150/MWh



Maximum load: 50 MW
Bid price: €50/MWh

Market outcomes:

- Productions of G1 and G2: **50 MW and 50 MW**
- Consumptions of D1 and D2: **100 MW and 0 MW**
- Market-clearing price: **€ 100 /MWh**

An Optimisation Model for the Electricity Market Clearing

An Optimisation Model for the Electricity Market Clearing

Objective Function (Generic form):

Maximize **social welfare (SW) of the market**¹

Subject to

- All technical constraints of generators and demands
- Power balance equality

¹ SW is equal to:

[total utility of demands based on their bid prices] – [total cost of generators based on their offer prices]

Optimisation model for MC: Simple Example

$$\text{Maximise } SW = \underbrace{\left[150p_1^D + 50p_2^D \right]}_{\text{Utility of demands}} - \underbrace{\left[40p_1^G + 100p_2^G \right]}_{\text{Cost of generators}} \quad (1a)$$

$\underbrace{p_1^D, p_2^D, p_1^G, p_2^G}_{\text{Set of primal variables}}$

subject to:

$$0 \leq p_1^D \leq 100 \quad (1b)$$

$$0 \leq p_2^D \leq 50 \quad (1c)$$

$$0 \leq p_1^G \leq 50 \quad (1d)$$

$$0 \leq p_2^G \leq 100 \quad (1e)$$

$$p_1^D + p_2^D - p_1^G - p_2^G = 0 \quad (1f)$$

(1b) } Consumption limits
 (1c) }
 (1d) } Generation limits
 (1e) }
 (1f) → Power balance

Optimisation model for MC

- **Question:**

How to obtain a market-clearing price within the optimization problem?

- **Answer:**

- ✓ The **dual variable** (also known as “Lagrangian multiplier”) of the power balance equality provides the market-clearing price (“uniform” pricing scheme)!
- ✓ It represents the marginal cost for an adjustment of the Power balance equality constraint

Optimisation model for MC: Simple Example

$$\underset{p_1^D, p_2^D, p_1^G, p_2^G}{\text{Maximise}} \quad SW = \left[150p_1^D + 50p_2^D \right] - \left[40p_1^G + 100p_2^G \right] \quad (1a)$$

subject to:

$$0 \leq p_1^D \leq 100 \quad (1b)$$

$$0 \leq p_2^D \leq 50 \quad (1c)$$

$$0 \leq p_1^G \leq 50 \quad (1d)$$

$$0 \leq p_2^G \leq 100 \quad (1e)$$

$$p_1^D + p_2^D - p_1^G - p_2^G = 0 \quad (1f)$$

$:\lambda$



Dual variable of power balance equality

Optimisation model for MC: Simple Example

$$\underset{p_1^D, p_2^D, p_1^G, p_2^G}{\text{Maximise}} \quad SW = [150p_1^D + 50p_2^D] - [40p_1^G + 100p_2^G] \quad (1a)$$

subject to:

$$0 \leq p_1^D \leq 100 \quad (1b)$$

$$0 \leq p_2^D \leq 50 \quad (1c)$$

$$0 \leq p_1^G \leq 50 \quad (1d)$$

$$0 \leq p_2^G \leq 100 \quad (1e)$$

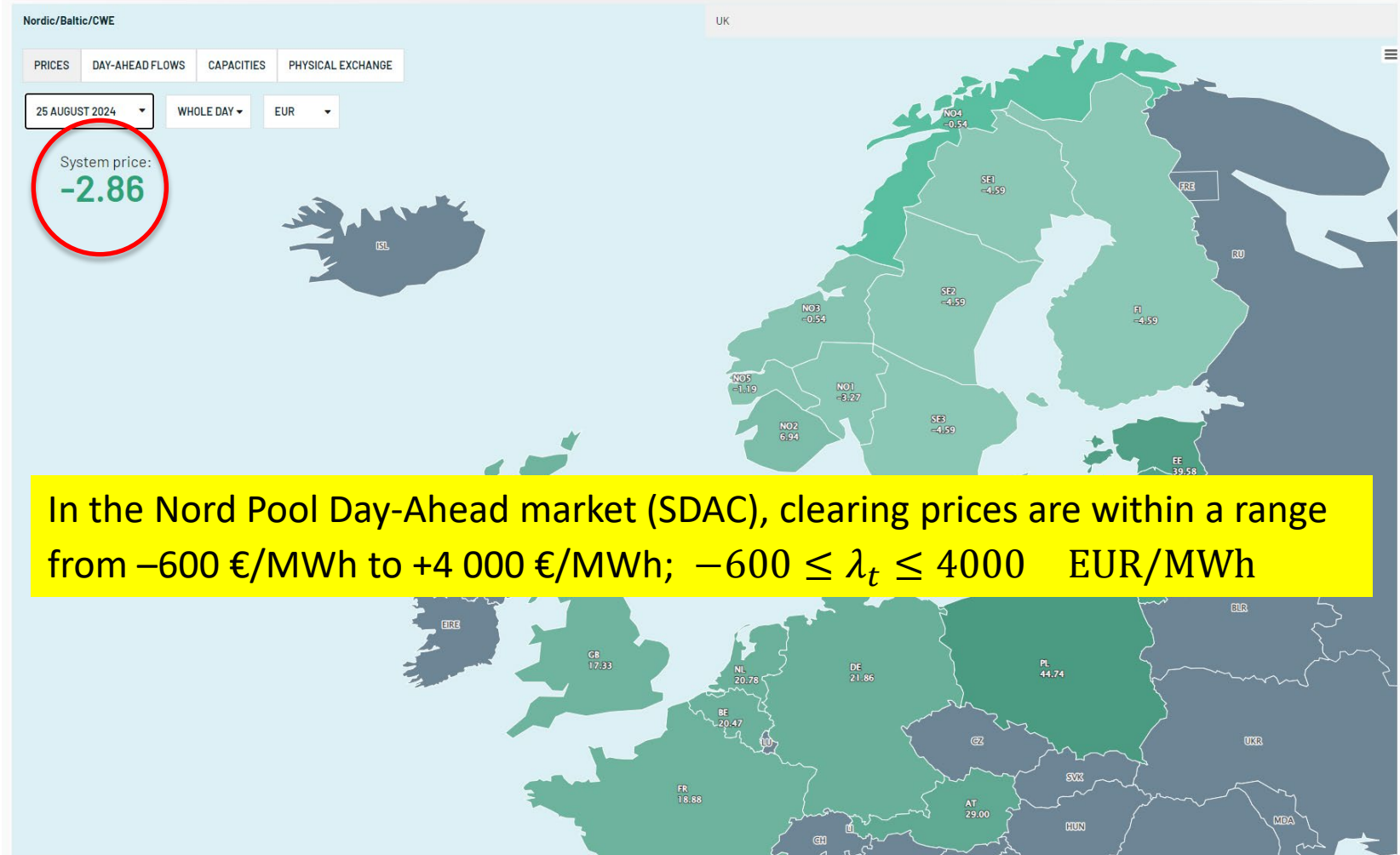
$$p_1^D + p_2^D - p_1^G - p_2^G = 0 \quad :\lambda \quad (1f)$$

Discussion:

- What does a dual variable show in general (mathematical interpretation)?
- What is its sign (negative, or positive, or free)? Can the electricity market price be negative?

Negative prices in the Nordic

Day-ahead overview



In the Nord Pool Day-Ahead market (SDAC), clearing prices are within a range from -600 €/MWh to +4 000 €/MWh; $-600 \leq \lambda_t \leq 4000$ EUR/MWh

Compact from-Introducing notations first

Inputs:

- All bids and offers in the market are formulated in terms of a *quantity* p and a *price* U_d and C_g for demand and supply, respectively.
- On the demand side (N_D demand bids):
 - set of bids: $P^D = \{p_d^D, d = 1, \dots, N_D\}$
 - maximum load of demands $p_d^D: \bar{P}_d^D$
 - bid prices for demand $p_d^D: U_d$
- On the supply side (N_G supply offers):
 - set of offers: $P^G = \{p_g^G, g = 1, \dots, N_G\}$
 - maximum capacity of generators $p_g^G: \bar{P}_g^G$
 - offer prices for generators $p_g^G: C_g$

Decision variables:

- *Consumption schedule*: $\mathbf{P}^D = [p_1^D, \dots, p_{N_D}^D]^T, 0 \leq p_d^D \leq \bar{P}_d^D$
- *Generation schedule*: $\mathbf{P}^G = [p_1^G, \dots, p_{N_G}^G]^T, 0 \leq p_g^G \leq \bar{P}_g^G$

Centralized social welfare optimization

- The social welfare maximization problem can be written as

$$\underset{p_g^G, p_d^D}{\text{maximise}} \quad SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G \quad (1a)$$

subject to:

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D \quad (1b)$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G \quad (1c)$$

$$\sum_{d=1}^{N_D} p_d^D - \sum_{g=1}^{N_G} p_g^G = 0 \quad (1d)$$

- And equivalently as a minimization problem by minimizing the opposite objective function, i.e.

$$\underset{p_g^G, p_d^D}{\text{minimise}} \quad -SW = \sum_{g=1}^{N_G} C_g p_g^G - \sum_{d=1}^{N_D} U_d p_d^D \quad (2a)$$

subject to

$$(1b)-(1d) \quad (2b)$$

How to derive Lagrangian function?

A linear program is in **standard form** if it is written as

$$\begin{array}{ll} \underset{x}{\text{Minimise}} & f(x) \\ \text{subject to:} & \\ h(x) = 0 & : \quad \lambda \\ g(x) \leq 0 & : \quad \mu \end{array}$$



$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T h(\mathbf{x}) + \boldsymbol{\mu}^T g(\mathbf{x})$$

<http://www.slimy.com/~steuard/teaching/tutorials/Lagrange.html>

How to derive optimality conditions?

Original (primal) problem

Minimise $f(x)$

subject to:

$$h(x) = 0 \quad : \quad \lambda$$

$$g(x) \leq 0 \quad : \quad \mu$$

Lagrangian function


$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T h(\mathbf{x}) + \boldsymbol{\mu}^T g(\mathbf{x})$$

Optimality Karush–Kuhn–Tucker (KKT) conditions


$$\frac{\partial \mathcal{L}(x, \lambda, \mu)}{\partial x} = 0$$

$$h(x) = 0$$

$$g(x) \leq 0$$

$$\mu \geq 0$$

$$\mu g(x) = 0$$

$$\lambda \in \text{free}$$

$$\frac{\partial \mathcal{L}(x, \lambda, \mu)}{\partial x} = 0$$

$$h(x) = 0$$

$$0 \leq -g(x) \perp \mu \geq 0$$

$$\lambda \in \text{free}$$

Complementarity
Slackness Condition

Simple Example

We convert the previous maximisation problem to minimisation problem

$$\text{Minimise } 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D$$

$$p_1^D, p_2^D, p_1^G, p_2^G$$

subject to:

Four primal variables

$$p_1^D + p_2^D - p_1^G - p_2^G = 0:$$

$$\lambda$$

$$p_1^D \leq 100:$$

$$\mu_1^D$$

$$p_2^D \leq 50:$$

$$\mu_2^D$$

$$p_1^G \leq 50:$$

$$\mu_1^G$$

$$p_2^G \leq 100:$$

$$\mu_2^G$$

$$p_1^D \geq 0:$$

$$\gamma_1^D$$

$$p_2^D \geq 0:$$

$$\gamma_2^D$$

$$p_1^G \geq 0:$$

$$\gamma_1^G$$

$$p_2^G \geq 0:$$

$$\gamma_2^G$$

Simple Example

We convert the previous problem to standard form

$$\text{Minimise } 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D$$
$$p_1^D, p_2^D, p_1^G, p_2^G$$

subject to:

$$p_1^D + p_2^D - p_1^G - p_2^G = 0:$$

 λ

$$p_1^D \leq 100:$$

 μ_1^D

$$p_2^D \leq 50:$$

 μ_2^D

$$p_1^G \leq 50:$$

 μ_1^G

$$p_2^G \leq 100:$$

 μ_2^G

$$-p_1^D \leq 0:$$

 γ_1^D

$$-p_2^D \leq 0:$$

 γ_2^D

$$-p_1^G \leq 0:$$

 γ_1^G

$$-p_2^G \leq 0:$$

 γ_2^G

Recall:

When we derive the Lagrangian function, the inequality constraints should be in the form of

$$g(x) \leq 0$$

Simple Example

Original (primal) problem

$$\underset{p_1^D, p_2^D, p_1^G, p_2^G}{\text{Minimise}} \quad \left[40 p_1^G + 100 p_2^G \right] - \left[150 p_1^D + 50 p_2^D \right]$$

subject to:

$$\begin{array}{ll} p_1^D + p_2^D - p_1^G - p_2^G = 0: & \lambda \\ p_1^D \leq 100: & \mu_1^D \\ p_2^D \leq 50: & \mu_2^D \\ p_1^G \leq 50: & \mu_1^G \\ p_2^G \leq 100: & \mu_2^G \\ -p_1^D \leq 0: & \gamma_1^D \\ -p_2^D \leq 0: & \gamma_2^D \\ -p_1^G \leq 0: & \gamma_1^G \\ -p_2^G \leq 0: & \gamma_2^G \end{array}$$

Lagrangian function

$$\begin{aligned} \mathcal{L}(p_{g,d}^{G,D}, \lambda, \mu_{g,d}^{G,D}, \gamma_{g,d}^{G,D}) = & \left[40 p_1^G + 100 p_2^G \right] - \left[150 p_1^D + 50 p_2^D \right] \\ & + \lambda \left[p_1^D + p_2^D - p_1^G - p_2^G \right] \\ & + \mu_1^D \left[p_1^D - 100 \right] + \mu_2^D \left[p_2^D - 50 \right] \\ & + \mu_1^G \left[p_1^G - 50 \right] + \mu_2^G \left[p_2^G - 100 \right] \\ & - \gamma_1^D p_1^D - \gamma_2^D p_2^D - \gamma_1^G p_1^G - \gamma_2^G p_2^G \end{aligned}$$

Simple Example

Lagrangian function

$$\begin{aligned}\mathcal{L}(p_{g,d}^{G,D}, \lambda, \mu_{g,d}^{G,D}, \lambda_{g,d}^{G,D}) = & 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D \\ & + \lambda[p_1^D + p_2^D - p_1^G - p_2^G] \\ & + \mu_1^D[p_1^D - 100] + \mu_2^D[p_2^D - 50] \\ & + \mu_1^G[p_1^G - 50] + \mu_2^G[p_2^G - 100] \\ & - \gamma_1^D p_1^D - \gamma_2^D p_2^D - \gamma_1^G p_1^G - \gamma_2^G p_2^G\end{aligned}$$

Optimality KKT conditions

We can solve KKT conditions as a system of equations!

We have tried that in TET4186 😊

$$\frac{\partial \mathcal{L}(\cdot)}{\partial p_1^G} = 40 - \lambda + \mu_1^G - \gamma_1^G = 0, \quad 0 \leq 100 - p_1^D \perp \mu_1^D \geq 0$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial p_2^G} = 100 - \lambda + \mu_2^G - \gamma_2^G = 0, \quad 0 \leq 50 - p_2^D \perp \mu_2^D \geq 0$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial p_1^D} = -150 + \lambda + \mu_1^D - \gamma_1^D = 0, \quad 0 \leq 50 - p_1^G \perp \mu_1^G \geq 0$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial p_2^D} = -50 + \lambda + \mu_2^D - \gamma_2^D = 0, \quad 0 \leq 100 - p_2^G \perp \mu_2^G \geq 0$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial \lambda} = p_1^D + p_2^D - p_1^G - p_2^G = 0, \quad 0 \leq p_1^D \perp \gamma_1^D \geq 0, 0 \leq p_2^D \perp \gamma_2^D \geq 0, 0 \leq p_1^G \perp \gamma_1^G \geq 0, 0 \leq p_2^G \perp \gamma_2^G \geq 0$$

How to derive **dual problem**?

How to derive **dual problem**?

Original (primal) problem

Minimise $f(x)$

subject to:

$h(x) = 0$: λ

$g(x) \leq 0$: μ



Lagrangian function

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \lambda^T h(\mathbf{x}) + \mu^T g(\mathbf{x})$$

Step 1: derive “dual function” as Minimize $\mathcal{L}(\mathbf{x}, \lambda, \mu)$
 $\mathbf{x}$

- Dual function is an unconstrained optimization problem. For arbitrarily given values of dual variables (λ & μ should be non-negative), the dual function minimizes the (relaxed) Lagrangian function. Primal variables are the only variables to be optimized.
- The optimal value of the dual function provides a “**lower bound**” for the optimal value of objective function of the original primal problem.
- More info? Watch this short video: <https://www.youtube.com/watch?v=4OifjG2kIJQ>

How to derive **dual problem**?

Original (primal) problem

Minimise $f(x)$

subject to:

$h(x) = 0$: λ

$g(x) \leq 0$: μ



Lagrangian function

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \lambda^T h(\mathbf{x}) + \mu^T g(\mathbf{x})$$

Step 2: derive **“dual problem”** which provides the best possible lower bound, i.e.,

$$\text{Maximise}_{\lambda \in \text{free}, \mu \geq 0} \underbrace{\text{Minimise}_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \mu)}$$

dual function (i.e., lower bound)

Simple Example (recall our previous example)

Original (primal) problem

$$\text{Minimise}_{p_1^D, p_2^D, p_1^G, p_2^G} 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D$$

subject to:

$$p_1^D + p_2^D - p_1^G - p_2^G = 0:$$

$$p_1^D \leq 100:$$

$$p_2^D \leq 50:$$

$$p_1^G \leq 50:$$

$$p_2^G \leq 100:$$

$$-p_1^D \leq 0:$$

$$-p_2^D \leq 0:$$

$$-p_1^G \leq 0:$$

$$-p_2^G \leq 0:$$

λ

μ_1^D

μ_2^D

μ_1^G

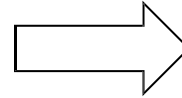
μ_2^G

γ_1^D

γ_2^D

γ_1^G

γ_2^G



Lagrangian function

$$\begin{aligned} \mathcal{L}(p_g^G, p_d^D, \lambda, \mu_g^G, \mu_d^D, \lambda_g^G, \lambda_d^D) = & 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D + \\ & \lambda[p_1^D + p_2^D - p_1^G - p_2^G = 0] \\ & + \mu_1^D[p_1^D - 100] + \mu_2^D[p_2^D - 50] \\ & + \mu_1^G[p_1^G - 50] + \mu_2^G[p_2^G - 100] \\ & - \gamma_1^D p_1^D - \gamma_2^D p_2^D - \gamma_1^G p_1^G - \gamma_2^G p_2^G \end{aligned}$$

These four terms are constants in the inner (i.e., minimization) problem, but variables in the outer (i.e., maximization) problem!

Dual problem

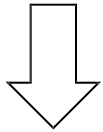
Maximise
 $\lambda \in \text{free}, (\mu_{1,2}^{D,G}, \gamma_{1,2}^{D,G}) \geq 0$

$$\left\{ \begin{aligned} & \text{Minimise}_{p_1^G, p_2^G, p_1^D, p_2^D} 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D \\ & + \lambda[p_1^D + p_2^D - p_1^G - p_2^G] \\ & + \mu_1^D[p_1^D - 100] + \mu_2^D[p_2^D - 50] \\ & + \mu_1^G[p_1^G - 50] + \mu_2^G[p_2^G - 100] \\ & - \gamma_1^D p_1^D - \gamma_2^D p_2^D - \gamma_1^G p_1^G - \gamma_2^G p_2^G \end{aligned} \right\}$$

Simple Example (recall our previous example)

Dual problem:

$$\text{Maximise}_{\lambda \in \text{free}, (\mu_{1,2}^{D,G}, \gamma_{1,2}^{D,G}) \geq 0} \left\{ \begin{array}{l} \text{Minimise}_{p_1^G, p_2^G, p_1^D, p_2^D} 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D \\ + \lambda [p_1^D + p_2^D - p_1^G - p_2^G] \\ + \mu_1^D [p_1^D \text{ (circled)} - 100] + \mu_2^D [p_2^D \text{ (circled)} - 50] \\ + \mu_1^G [p_1^G \text{ (circled)} - 50] + \mu_2^G [p_2^G \text{ (circled)} - 100] \\ - \gamma_1^D p_1^D - \gamma_2^D p_2^D - \gamma_1^G p_1^G - \gamma_2^G p_2^G \end{array} \right\}$$



$$\text{Maximise}_{\lambda \in \text{free}, (\mu_{1,2}^{D,G}, \gamma_{1,2}^{D,G}) \geq 0} -100\mu_1^D - 50\mu_2^D - 50\mu_1^G - 100\mu_2^G$$

subject to:

$$40 - \lambda + \mu_1^G - \gamma_1^G = 0$$

$$100 - \lambda + \mu_2^G - \gamma_2^G = 0$$

$$-150 + \lambda + \mu_1^D - \gamma_1^D = 0$$

$$-50 + \lambda + \mu_2^D - \gamma_2^D = 0$$

We can write the dual problem in a more compact way

Note:

- Dual variables γ are isolated, since they do not appear in the objective function, and do not link constraints!
- We also know that they are non-negative.
- So, we can get rid of them by converting equalities to inequalities.

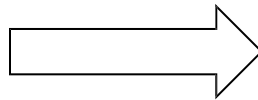
Simple Example (recall our previous example)

Dual problem:

$$\text{Maximise}_{\lambda \in \text{free}, (\mu_1^{D,G}, \gamma_1^{D,G}) \geq 0} -100\mu_1^D - 50\mu_2^D - 50\mu_1^G - 100\mu_2^G$$

subject to:

$$\begin{aligned} 40 - \lambda + \mu_1^G - \gamma_1^G &= 0 \\ 100 - \lambda + \mu_2^G - \gamma_2^G &= 0 \\ -150 + \lambda + \mu_1^D - \gamma_1^D &= 0 \\ -50 + \lambda + \mu_2^D - \gamma_2^D &= 0 \end{aligned}$$



$$\text{Maximise}_{\lambda \in \text{free}, (\mu_1^{D,G}) \geq 0} -100\mu_1^D - 50\mu_2^D - 50\mu_1^G - 100\mu_2^G$$

subject to:

$$\begin{aligned} 40 - \lambda + \mu_1^G &\geq 0 \\ 100 - \lambda + \mu_2^G &\geq 0 \\ -150 + \lambda + \mu_1^D &\geq 0 \\ -50 + \lambda + \mu_2^D &\geq 0 \end{aligned}$$

Simple Example (recall our previous example)

Original (primal) problem

$$\text{Minimise } 40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D$$

$$p_1^D, p_2^D, p_1^G, p_2^G \geq 0$$

subject to:

$$p_1^D + p_2^D - p_1^G - p_2^G = 0:$$

$$p_1^D \leq 100$$

$$p_2^D \leq 50$$

$$p_1^G \leq 50$$

$$p_2^G \leq 100$$

$$\lambda$$

$$\mu_1^D$$

$$\mu_2^D$$

$$\mu_1^G$$

$$\mu_2^G$$

Dual problem:

$$\text{Maximise } -100\mu_1^D - 50\mu_2^D - 50\mu_1^G - 100\mu_2^G$$

$$\lambda \in \text{free}, (\mu_{1,2}^{D,G}) \geq 0$$

subject to:

$$40 - \lambda + \mu_1^G \geq 0$$

$$100 - \lambda + \mu_2^G \geq 0$$

$$-150 + \lambda + \mu_1^D \geq 0$$

$$-50 + \lambda + \mu_2^D \geq 0$$

Important points

- ✓ Number of **variables** in the **primal** problem = Number of **constraints** in the **dual** problem
- ✓ Number of **constraints** in the **primal** problem = Number of **variables** in the **dual** problem
- ✓ Dual problem of the dual problem is the primal problem!
- ✓ Dual variables of the dual problem are the primal variables!

The dual problem

- Duality in linear programming is a very useful concept.
- Switching rules for primal and dual variables

Primal		Dual	
Objective	$Max\ z$	$Min\ z$	Objective
Variable (n)	≥ 0	≥ 0	Constraints (n)
	≤ 0	≤ 0	
	Unlimited	$=0$	
Constraints (m)	≤ 0	≥ 0	Variable (m)
	≥ 0	≤ 0	
	$=0$	Unlimited	
Right-side vector of constraints (b)		Coefficient vector of variables in objective function	
Coefficient vector of variables in objective function (c)		Right-side vector of constraints	

The dual problem, General notation (Recap from TET4185)

Consider the primal problem.

Primal Problem

Maximise $c^T x$
 $x \in \mathbb{R}^n$

s.t. $Ax \leq b \quad (\lambda)$

$x \geq 0.$

Then the dual problem is

Dual Problem

Minimise $b^T \lambda$
 $\lambda \in \mathbb{R}^n$

s.t. $A^T \lambda \geq c \quad (x)$

$\lambda \geq 0.$

Weak duality

The weak duality theorem states that

- If $\mathbf{x} \geq 0$ satisfies $\mathbf{Ax} \leq \mathbf{b}$
- and $\boldsymbol{\lambda} \geq 0$ satisfies $\mathbf{A}^T \boldsymbol{\lambda} \geq \mathbf{c}$
- then $\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \boldsymbol{\lambda}$

Hence, any feasible variable of the dual problem provides an upper bound for the primal problem (and vice versa). Why?

Proof

Given feasible $\mathbf{x}$ and $\boldsymbol{\lambda}$

$$\mathbf{c}^T \mathbf{x} \leq (\mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{x} = \boldsymbol{\lambda}^T \mathbf{Ax} \leq \boldsymbol{\lambda}^T \mathbf{b} = \mathbf{b}^T \boldsymbol{\lambda}$$

Strong duality

The strong duality theorem states that

- If there exists an optimal solution x^* to the primal problem
- then there exists an optimal solution, λ^* , to the dual problem
- such that $c^T x^* = b^T \lambda^*$

Important points

✓ Weak duality theorem:

- The value of objective function of the primal problem at any point of its feasible region is lower than or equal to that of the dual problem at any point of its feasible region.

- In our example:

$$\underbrace{40p_1^G + 100p_2^G - 150p_1^D - 50p_2^D}_{\text{The value of the objective function of the primal problem}} \leq \underbrace{-100\mu_1^D - 50\mu_2^D - 50\mu_1^G - 100\mu_2^G}_{\text{The value of the objective function of the dual problem}}$$

Important points

✓ **Strong duality theorem:**

In the optimal point (linear problems), the value of objective function of the dual problem is equal to that of the primal problem.

In our example [note that superscript * denotes the optimal value]:

$$\underbrace{40p_1^{G*} + 100p_2^{G*} - 150p_1^{D*} - 50p_2^{D*}}_{\text{The value of the objective function of the primal problem}} \overset{\text{red circle}}{=} \underbrace{-100\mu_1^{D*} - 50\mu_2^{D*} - 50\mu_1^{G*} - 100\mu_2^{G*}}_{\text{The value of the objective function of the dual problem}}$$

Exit ticket: lecture learning outcomes

Before leaving, check that you can answer these five questions without looking at the slides:

- 1 Write the social-welfare objective as utility minus generation cost.
- 2 Identify the power balance and capacity constraints in a market-clearing LP.
- 3 Explain why the dual variable of power balance is the uniform market price.
- 4 State weak and strong duality in your own words.
- 5 Preview next lecture: what changes when demand, prices or inflow are uncertain?

Take-home message: market clearing quantities and prices are two views of the same optimisation problem.